NumPy → Numerical Python

→ salary_list = [20000, 30k, 40k]

```
new_salary = []
for s in salary_list:
    new_salary.append(s * 1.10)
print(new_salary)
```

→ Hive
→ Spark SQL
↓
Data lake

Numpy
↓
C-level optimized computation
No python loop
work on entire data in a single go.

powerful object
↓
array
↓
mathematical operations.
pandas
Scikit learn
Tensor flow.

	list	Numpy
Data type	Mixed	Single type
Speed	Slow	fast
ML usage	X	✓

$$\begin{array}{c} \\ \rightarrow \\ \rightarrow \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \end{array}$$